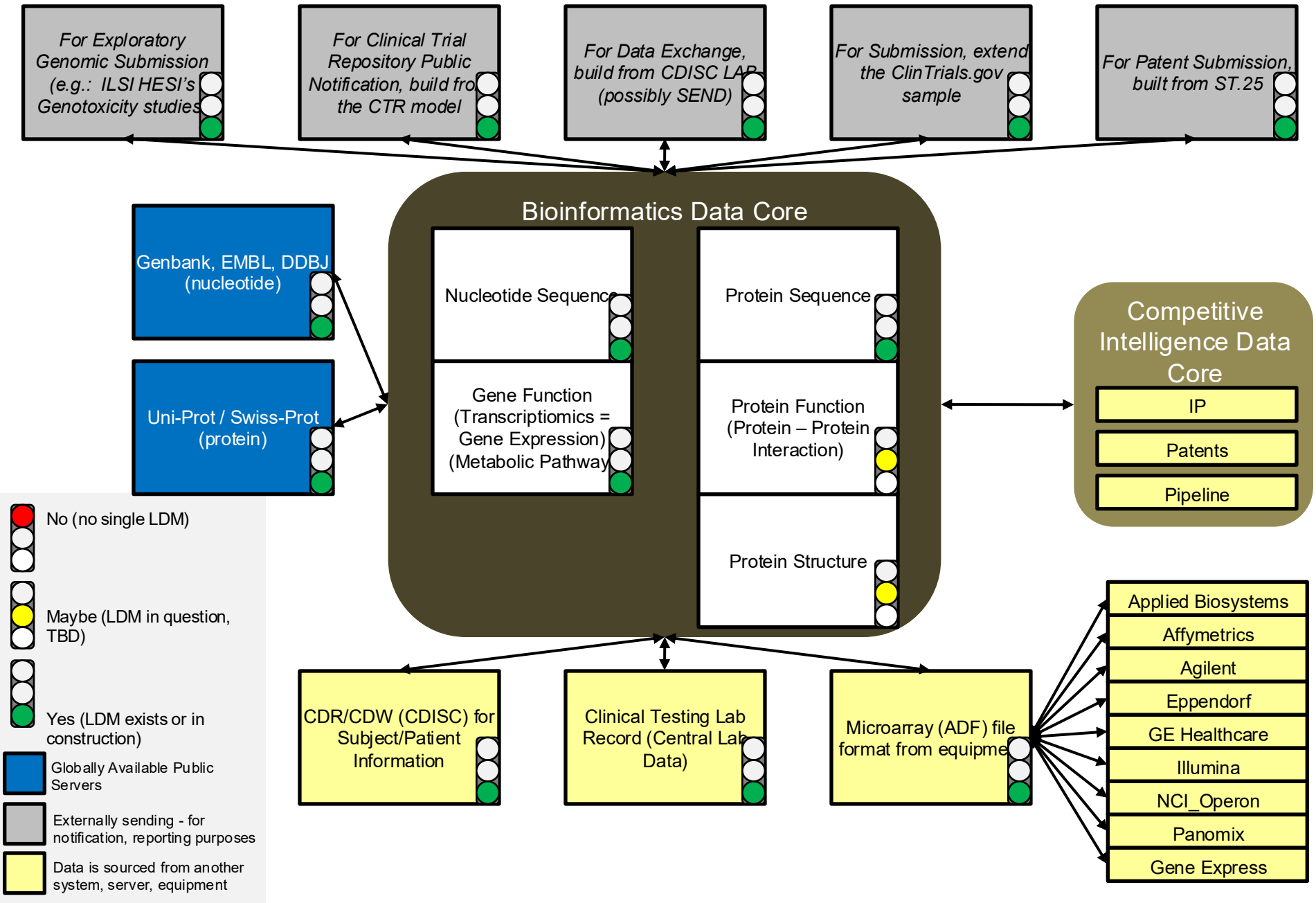
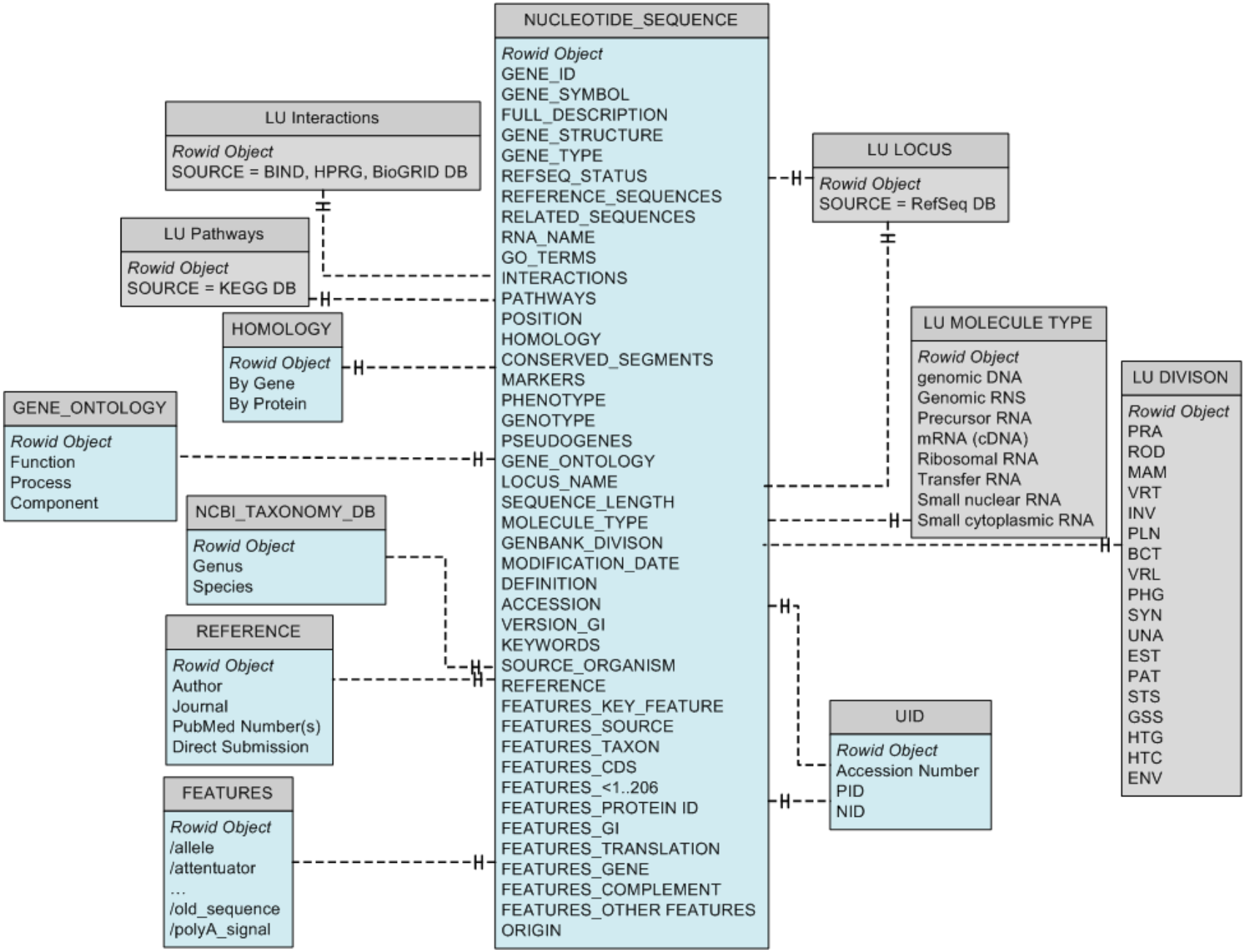
Overall Data "BLOBS" for Genomic services and operations



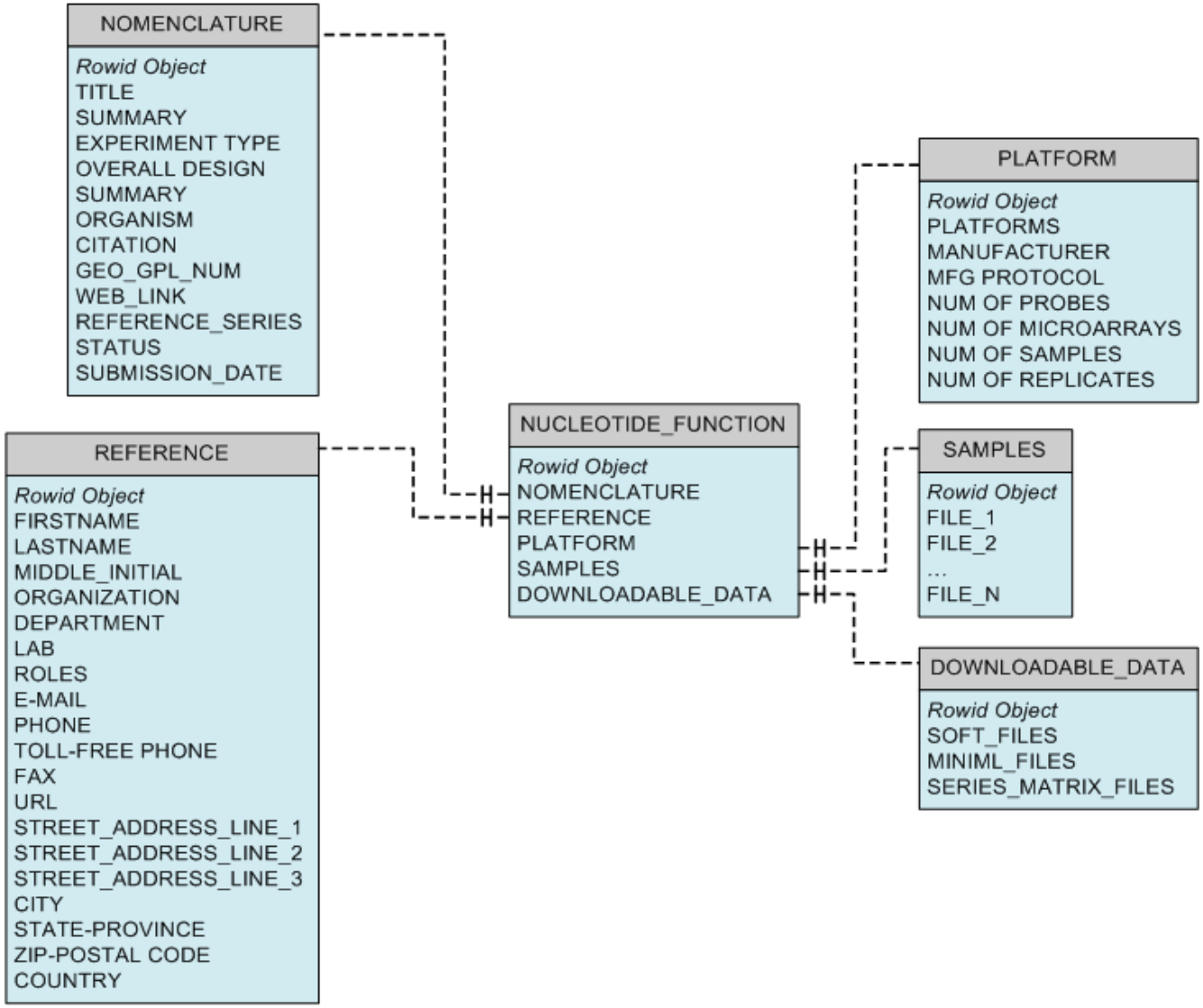
Logical Data Models

Nucleotide Sequence – Logical Data Model



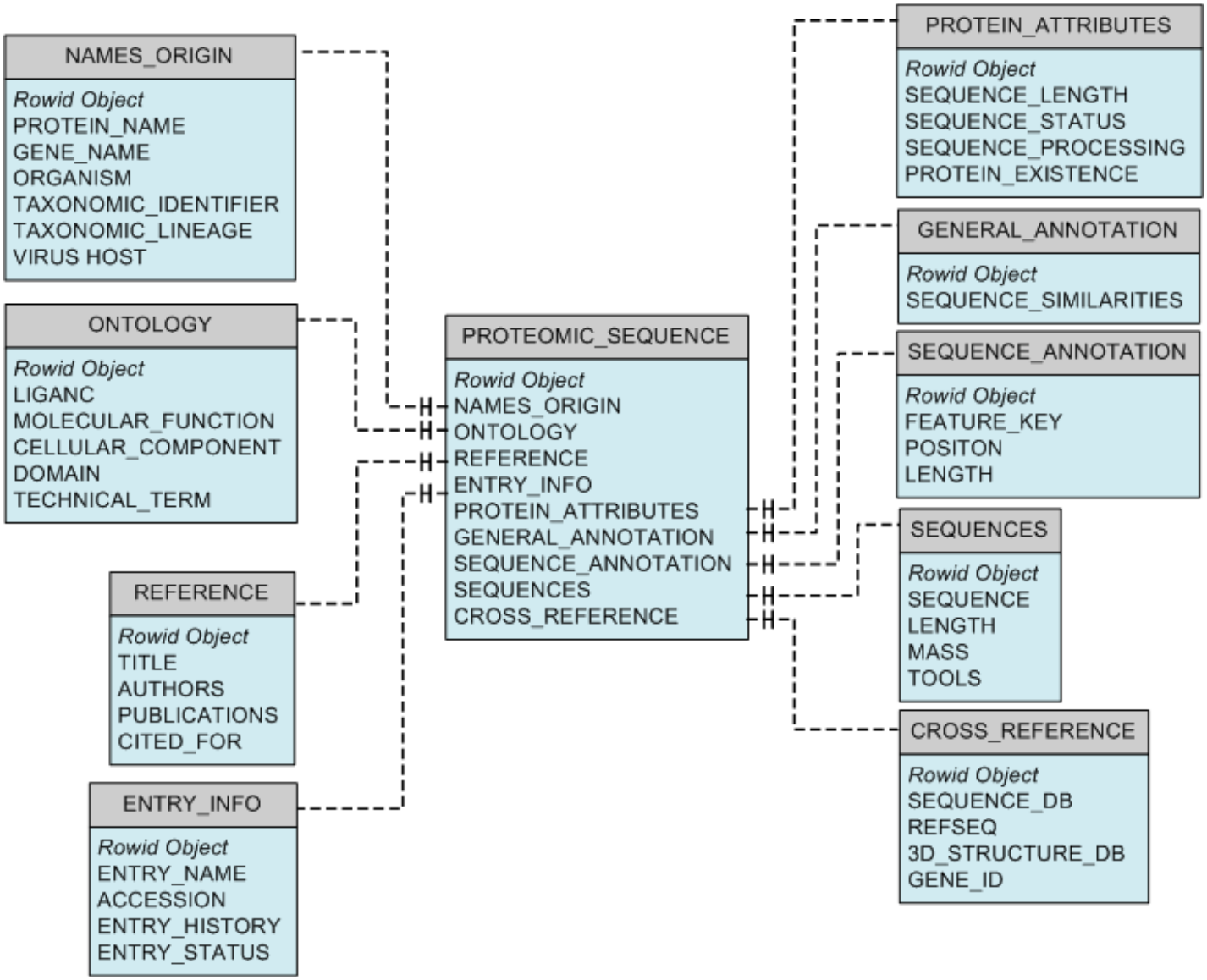
Nucleotide Function – Logical Data Model

A high-level construct with associated sub tables and look up tables to other sources



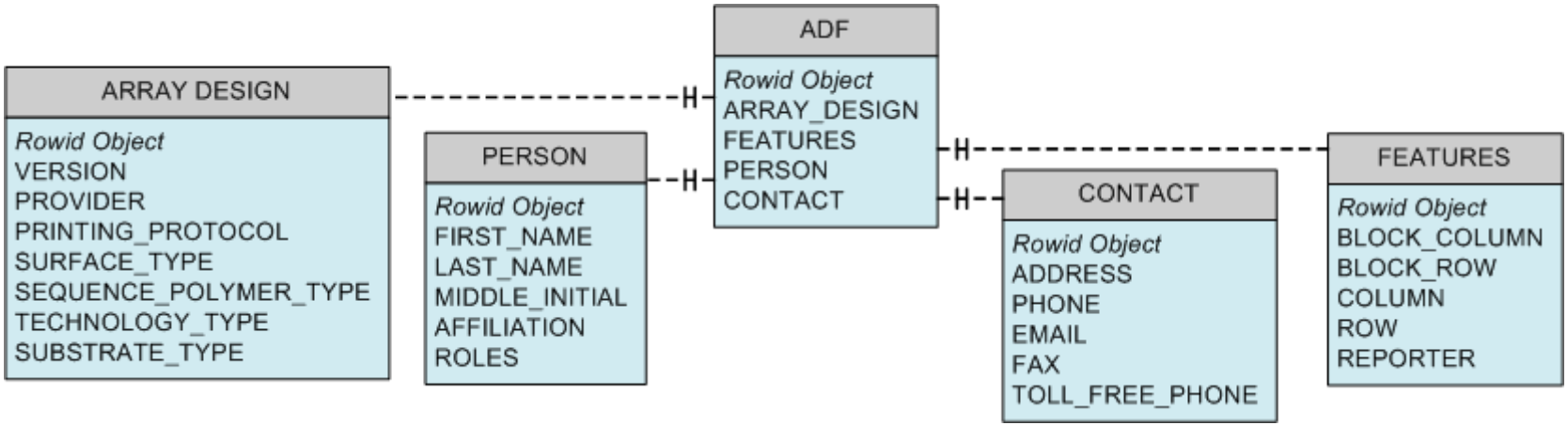
Protein Sequence – Logical Data Model

A high-level construct with associated sub tables and look up tables to other sources



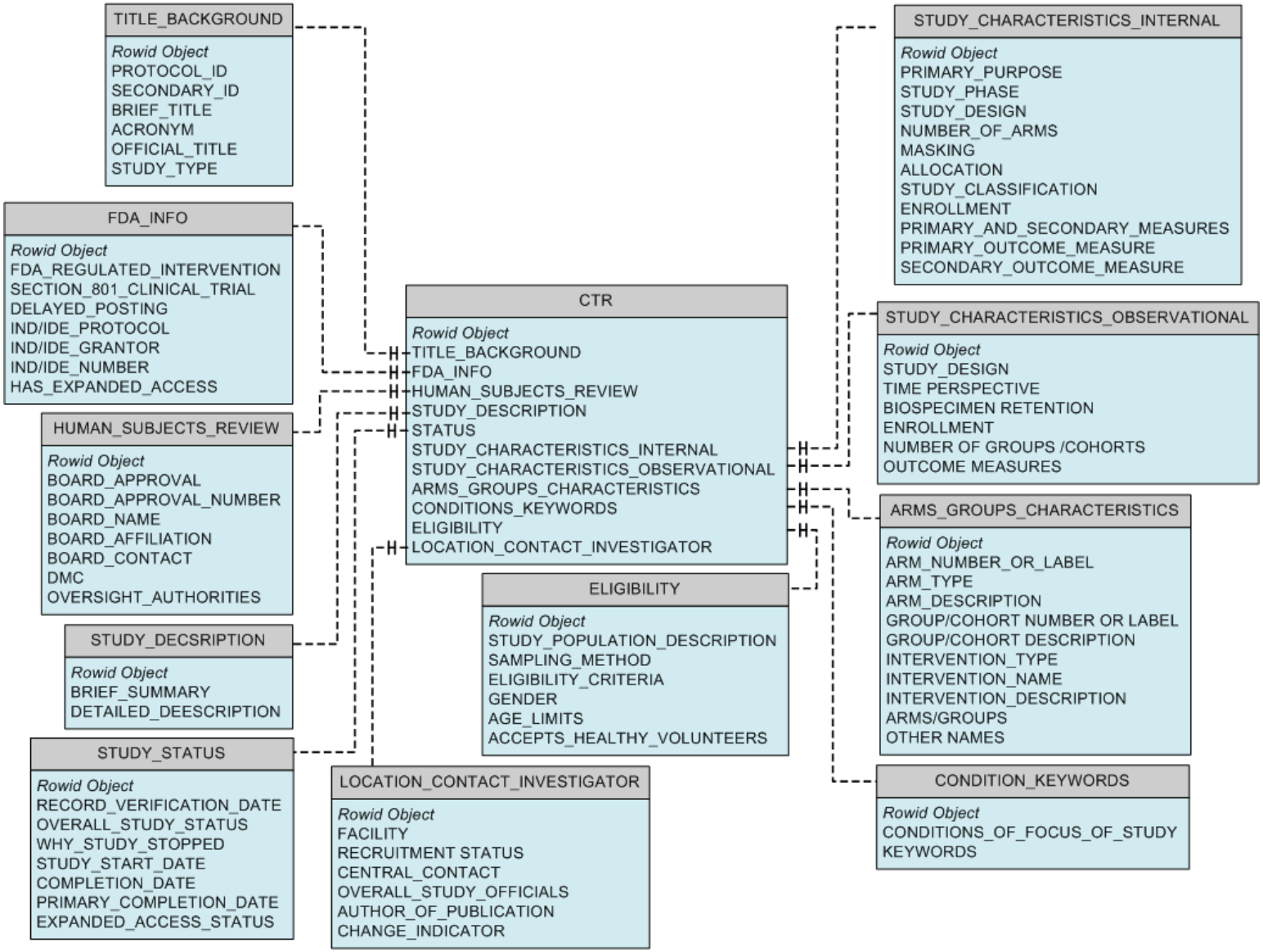
MicroArray Description Format – Logical Data Model

A high-level construct with associated sub tables and look up tables to other sources

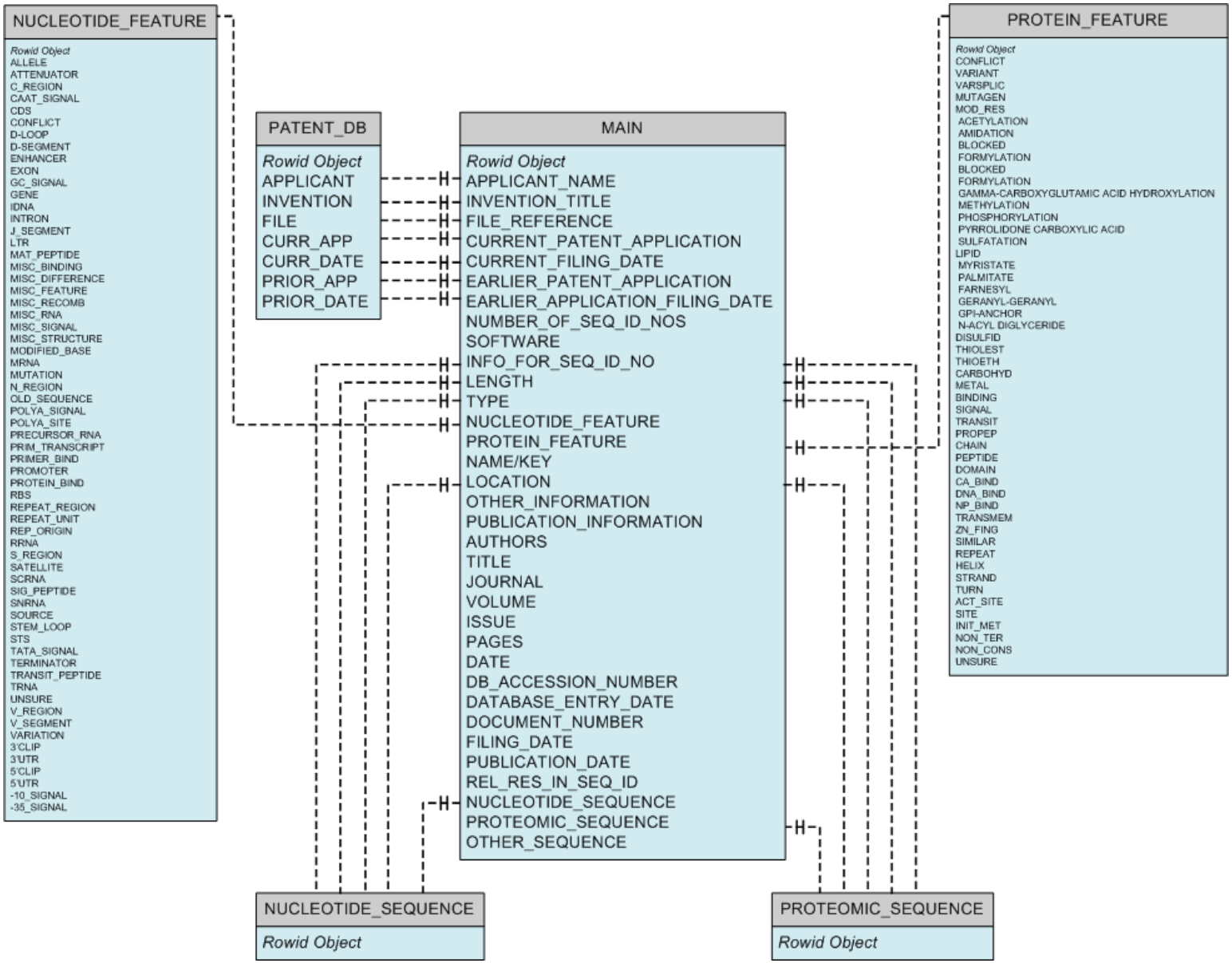


Clinical Trial Registry – Logical Data Model

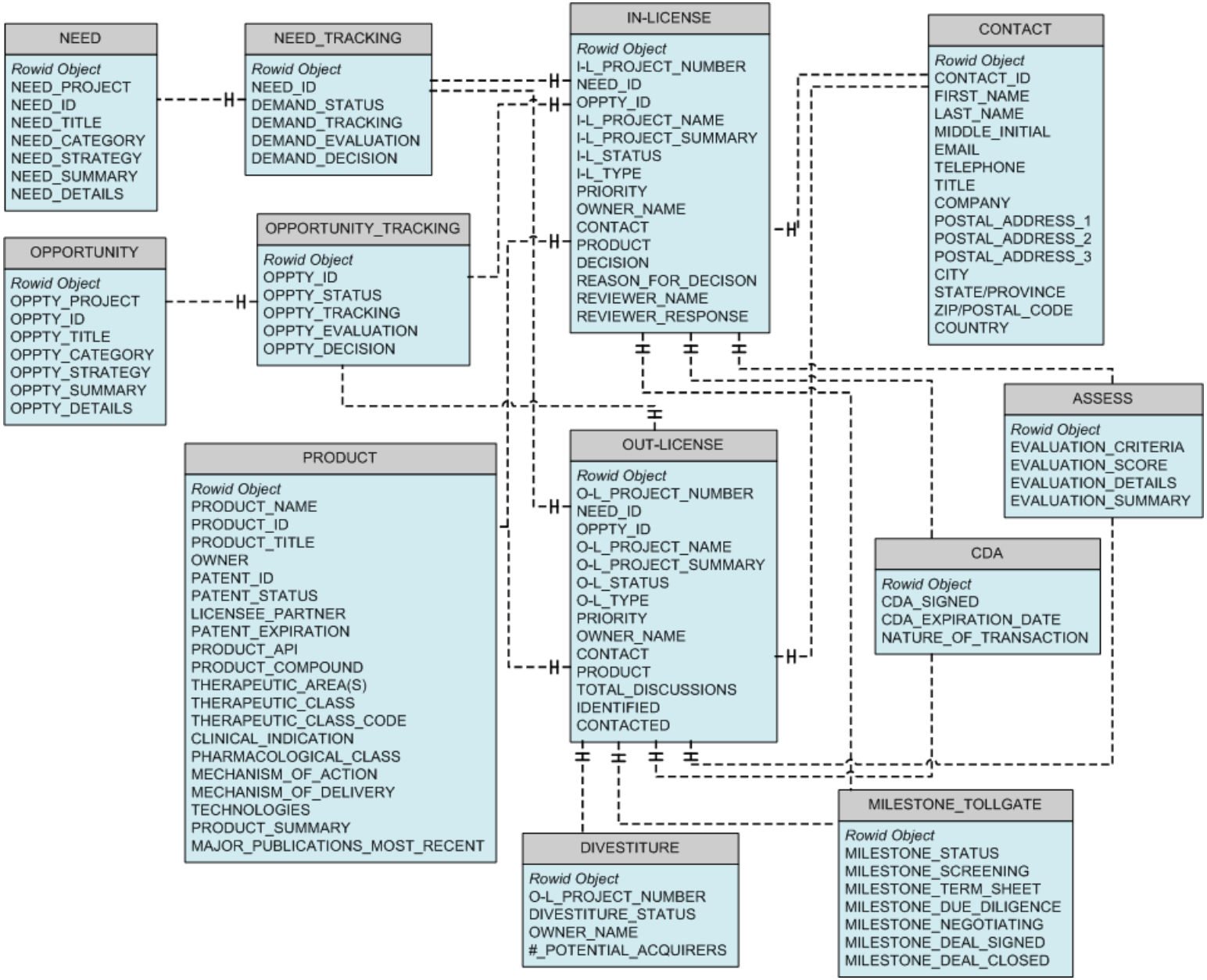
A high-level construct with associated sub tables and look up tables to other sources



A high-level construct with associated sub tables and look up tables to other sources

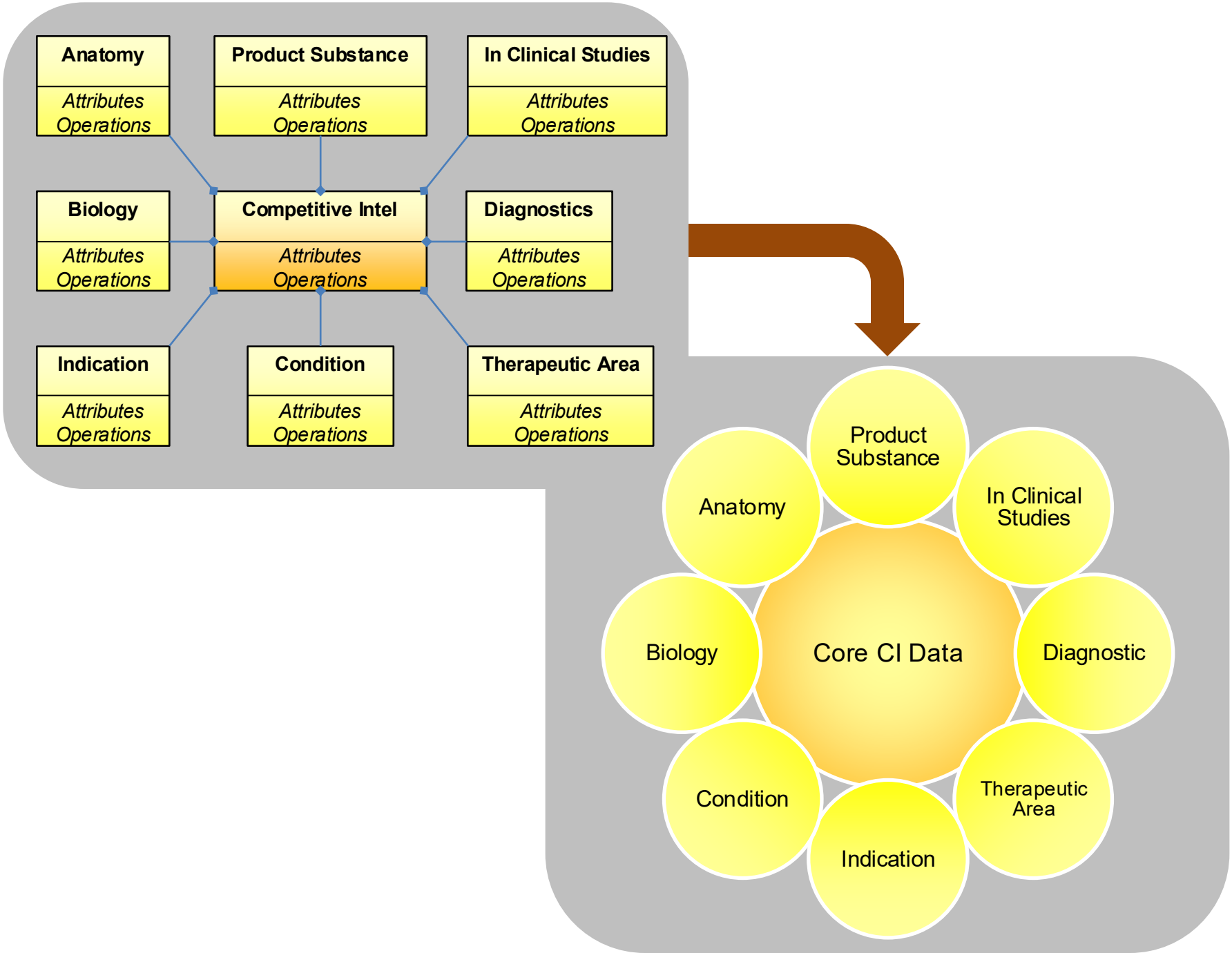


Business Development and Licensing – Logical Data Model

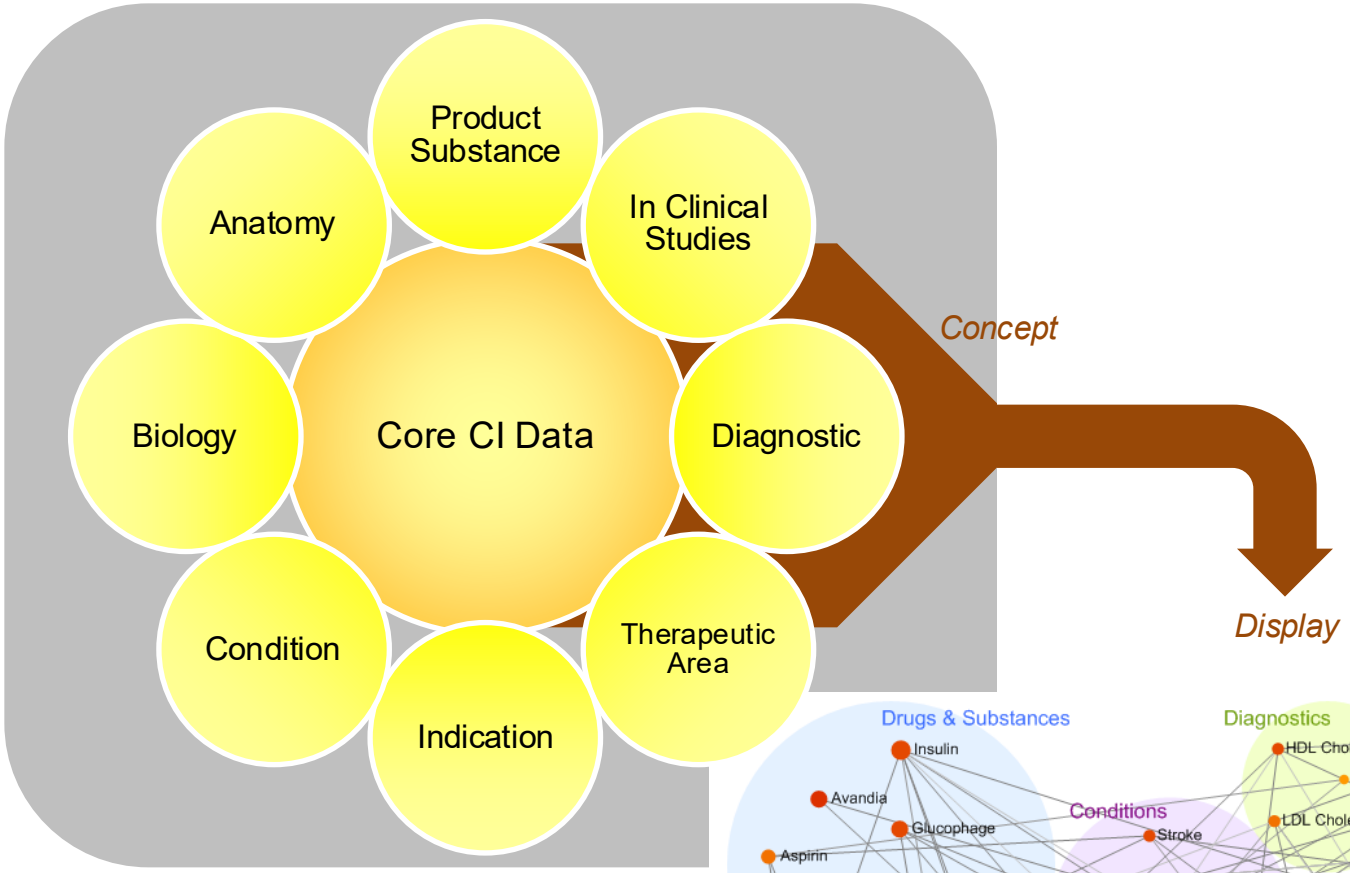


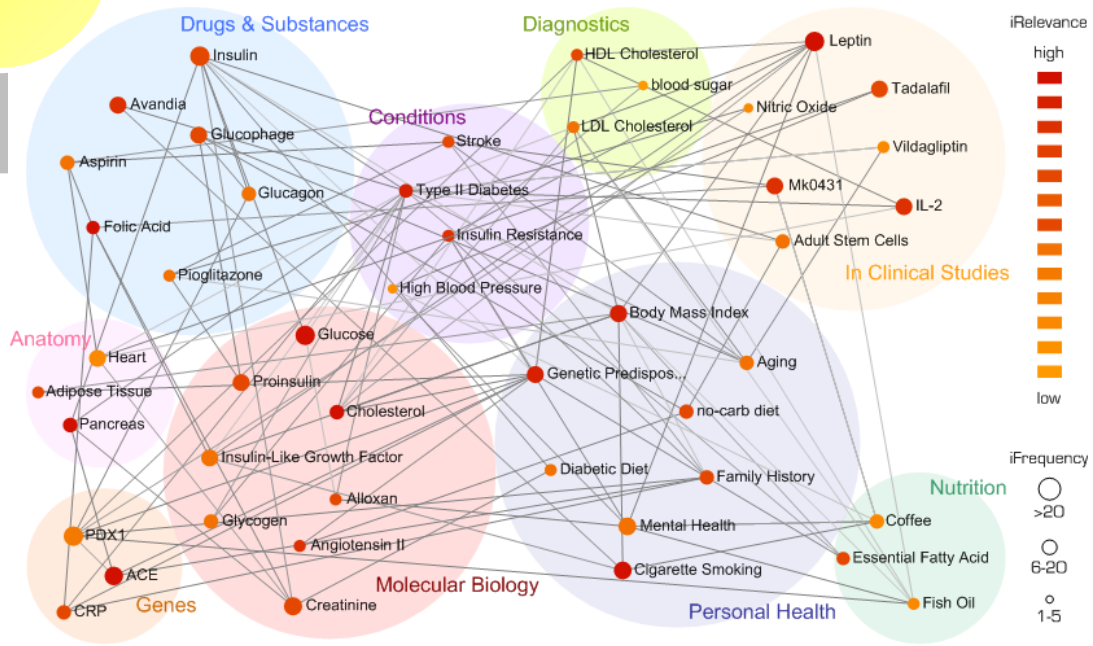
Conceptual Models

Competitive Intelligence – Core with Conceptual Domains

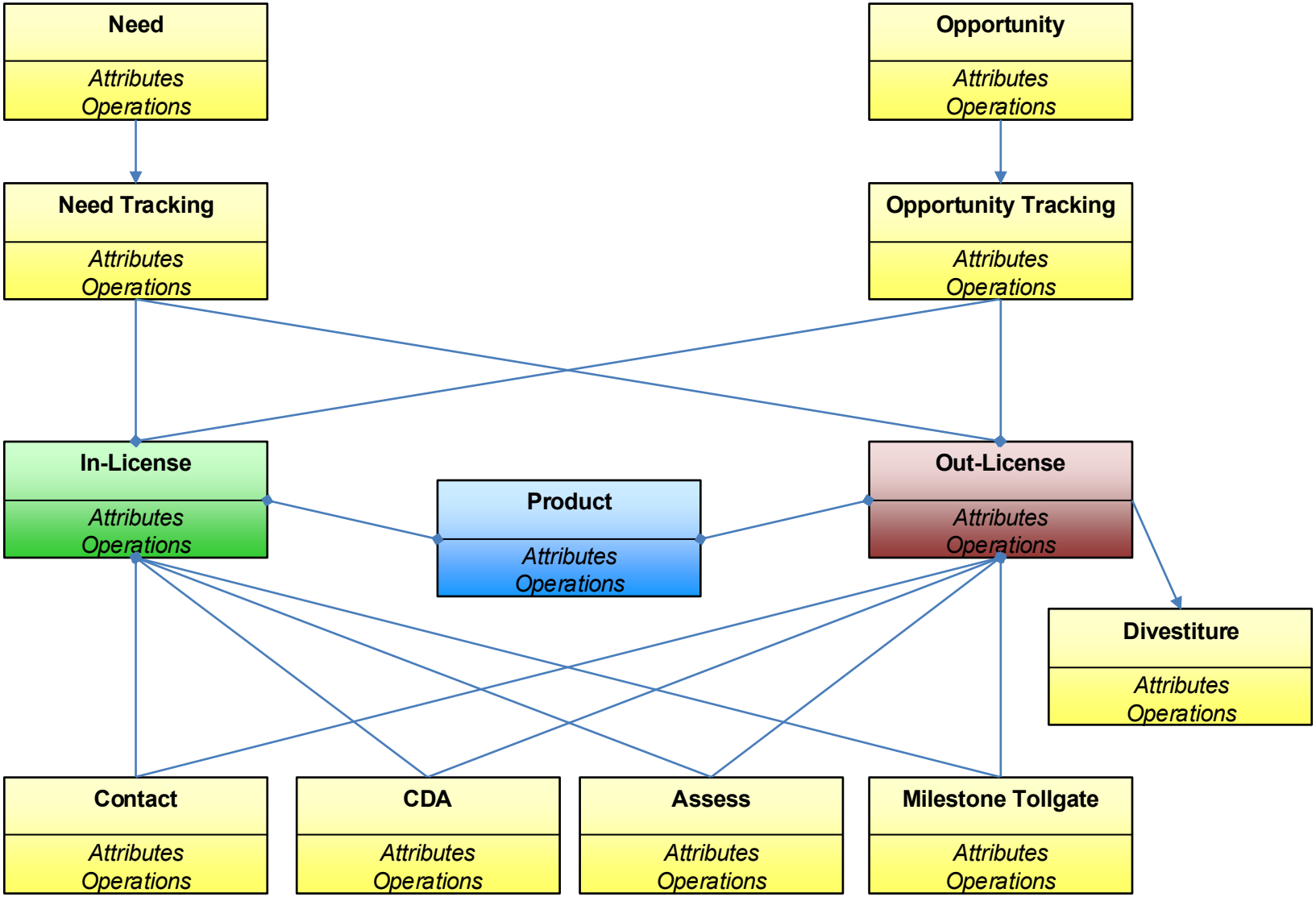


Competitive Intelligence – Conceptual Domain Shown in a Visual Discovery Mode

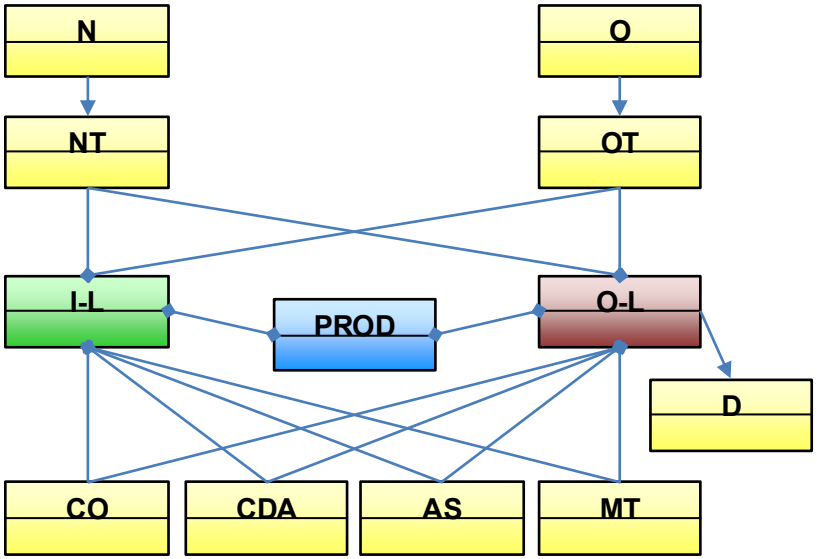




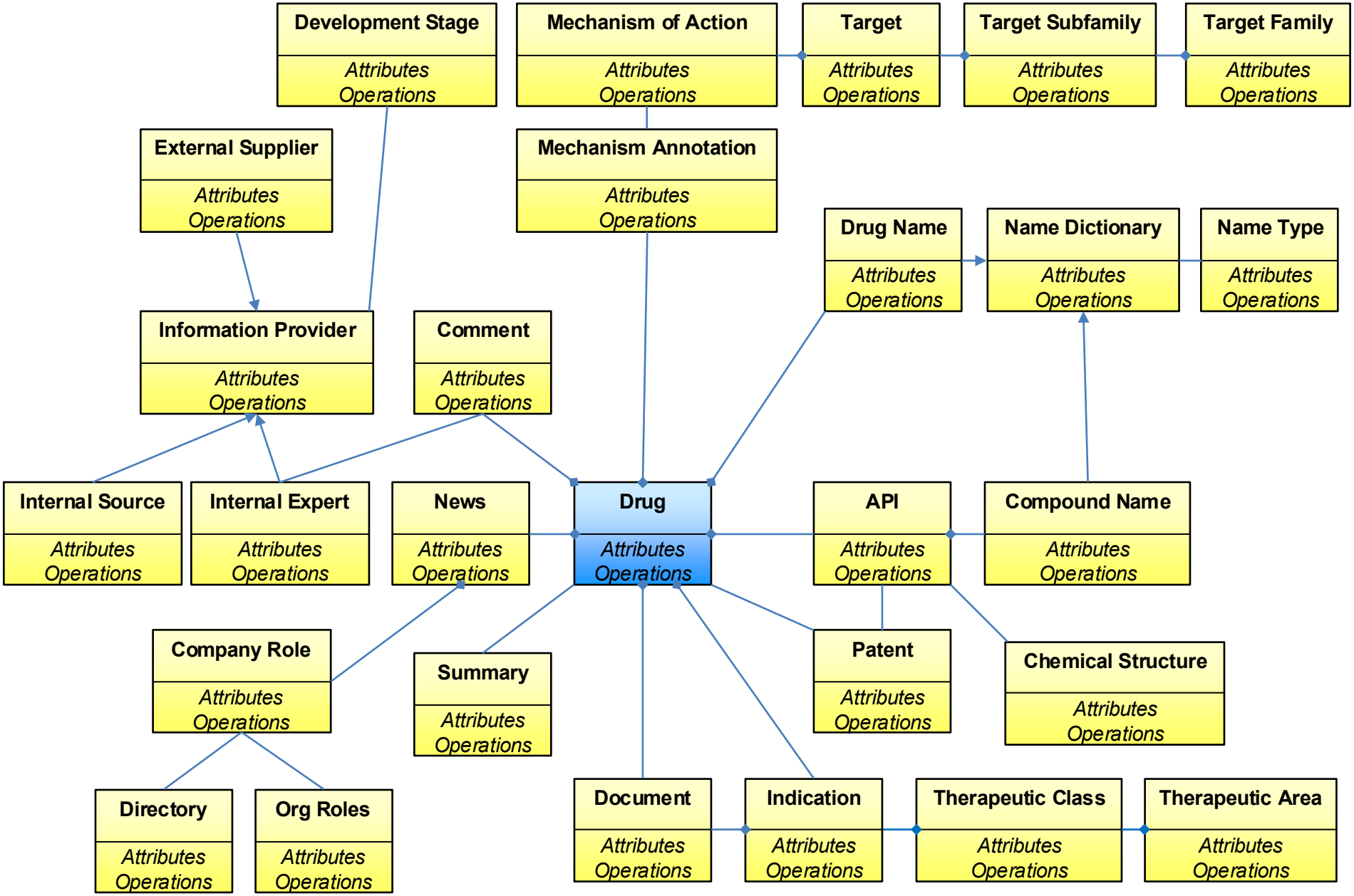
Business Development and Licensing – Conceptual Model



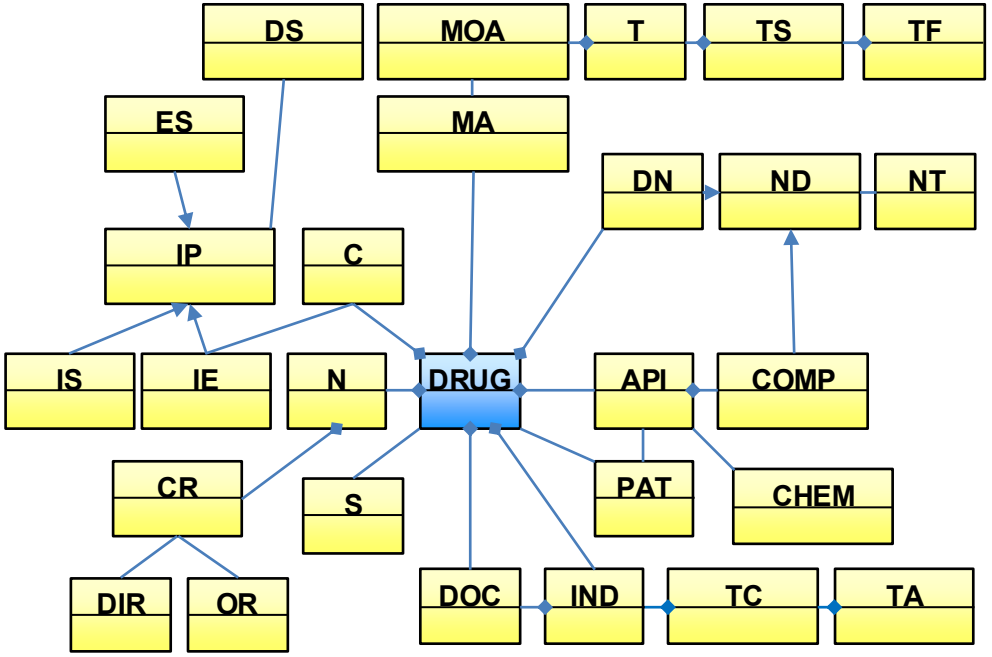
Business Development and Licensing – Conceptual Model (Miniaturized)



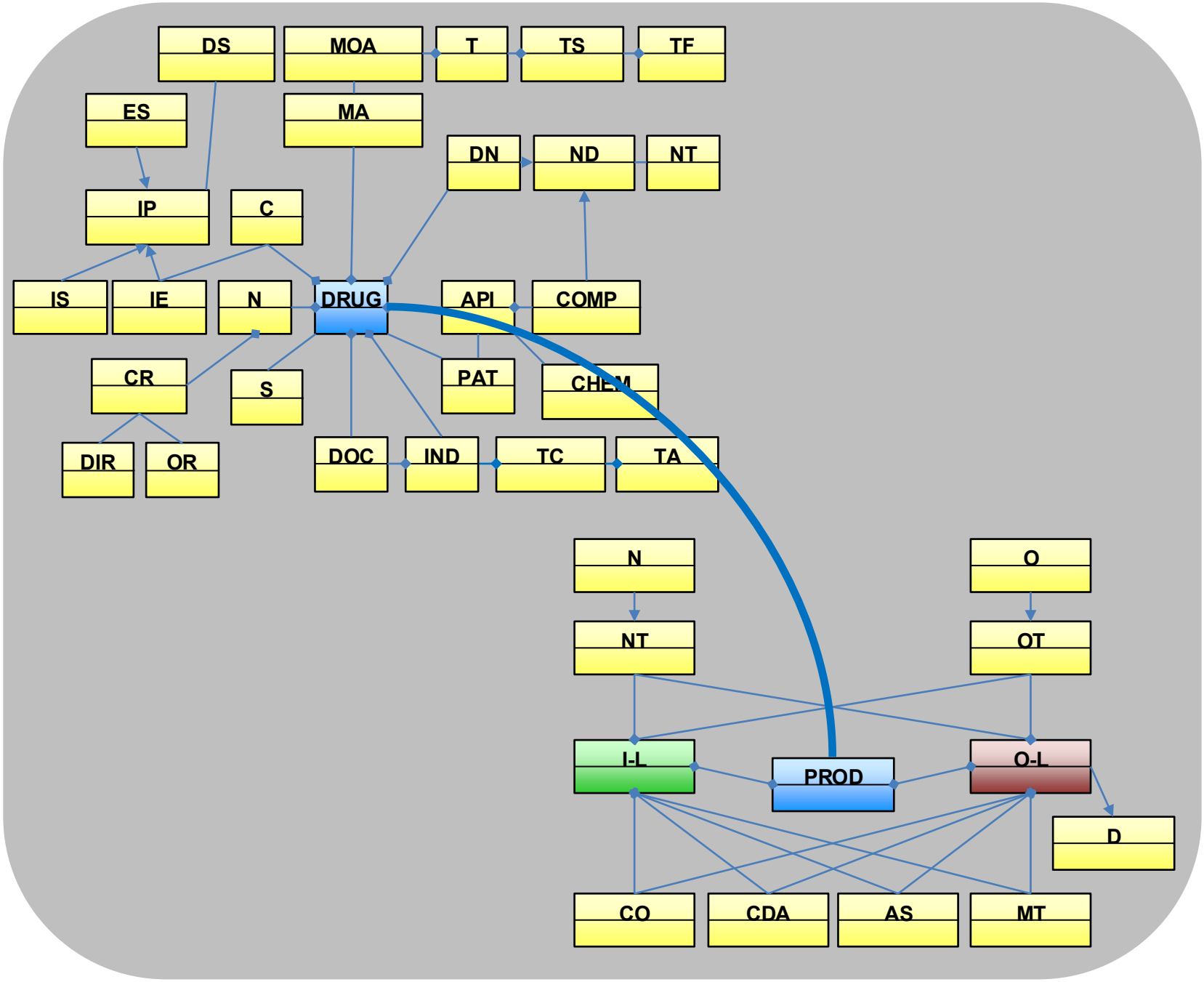
Competitive Intelligence – (ACCL High Level Class Diagram) Conceptual Model



Competitive Intelligence – ACCL HLCS Conceptual Model (Miniaturized)

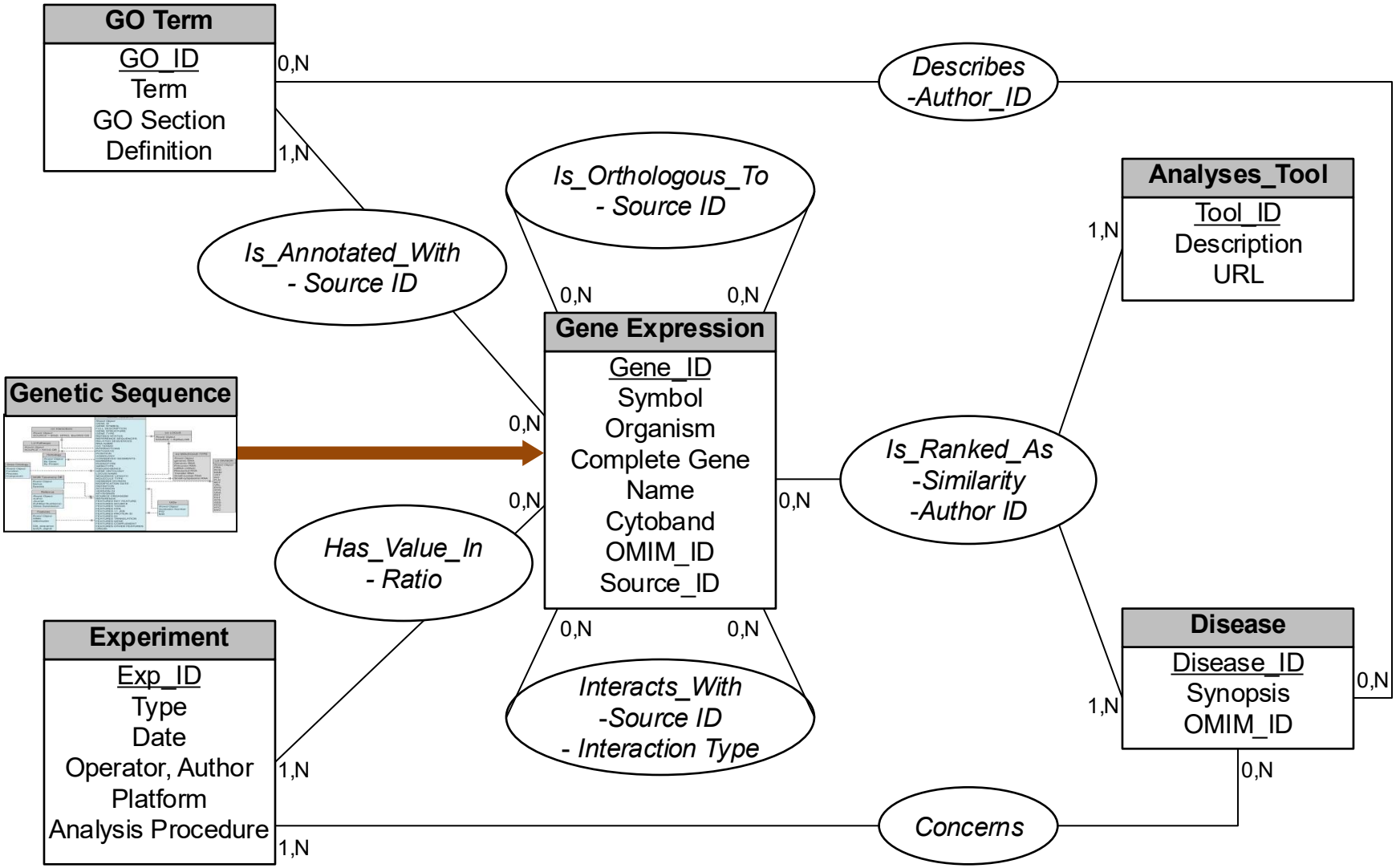


Competitive Intelligence – Link/Associate ACCL HLCS and CSC BD&L CI Concepts



Nucleotide Function (Genetic Expression Analyses) – Conceptual Data Model

For interactive/functional research, smaller constructs can be extended from the larger schema



Adapted from: Yilmaz, S. et al. Bioinformatics 2009 25:230-236; doi:10.1093/bioinformatics/btn612

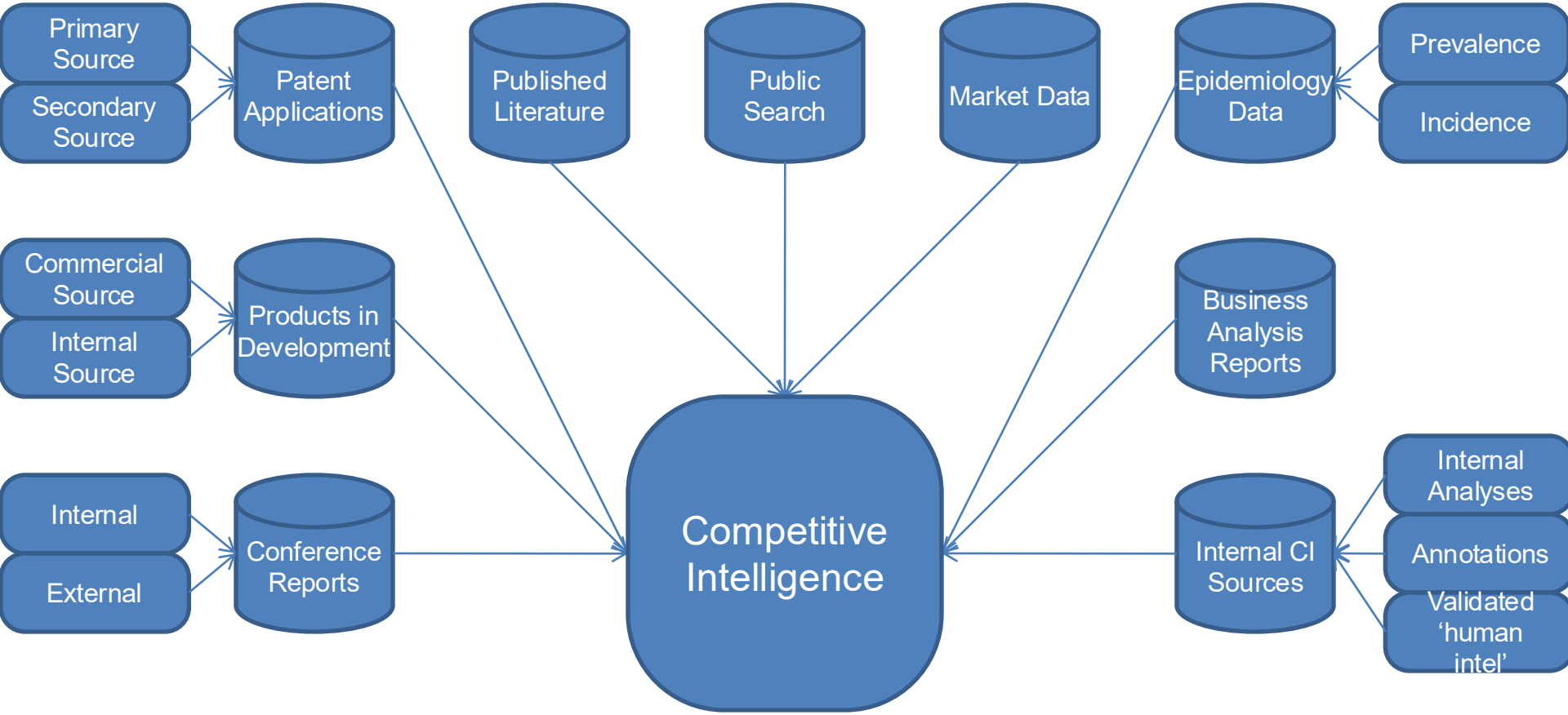
Genetic Expression Analyses – Logical Data Model (Table Schema)

The previously visualized conceptual model can also be put into a logical data model. Each table contains a set of attributes including a primary key (in bold face) and one or more foreign keys (in italics).

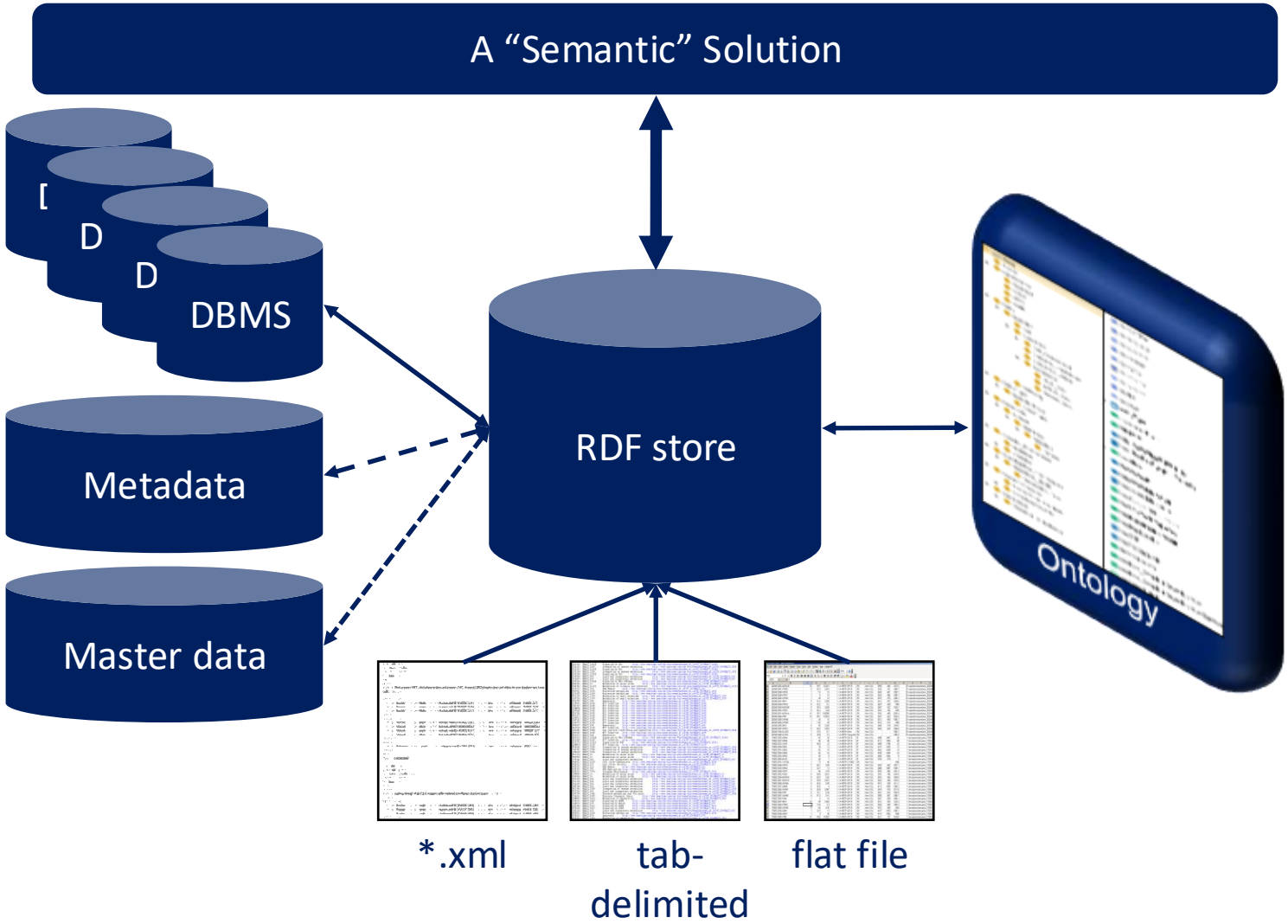
Gene	**Gene_ID**, *Symbol, Organism, Complete_name, Chromosome, Cytoband, OMIM_ID, Source_ID*
GO_Term	**GO_ID**, *Term, GO_section, Definition*
Gene_GO_Term	**Gene_ID, GO_ID**, *Source_ID*
Orthology	**Gene_ID1, Gene_ID2**, *Source_ID*
Interaction	**Gene_ID1, Gene_ID2**, *Source_ID, Interaction_Type*
Disease	**Disease_ID**, *Synopsis, OMIM_ID*
Disease_GO_Term	**Disease_ID, GO_ID, Author_ID**
Analyses_Tool	**Tool_ID**, *Description, URL*
Gene_Disease_Similarity	**Gene_ID, Disease_ID, Author_ID, Tool_ID**,*Similarity*
Experiment	**Exp_ID**, *Type, Date, Platform, Operator/Author, Analysis_Procedure, Disease_ID*
Gene_In_Experiment	**Gene_ID, Exp_ID**, *Ratio*

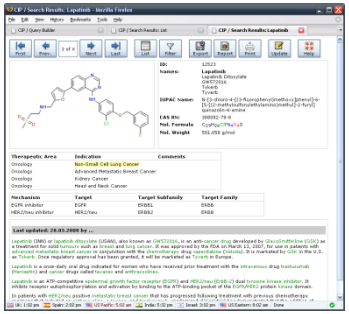
Adapted from: Yilmaz, S. et al. Bioinformatics 2009 25:230-236; doi:10.1093/bioinformatics/btn612

Competitive Intelligence Data Sources

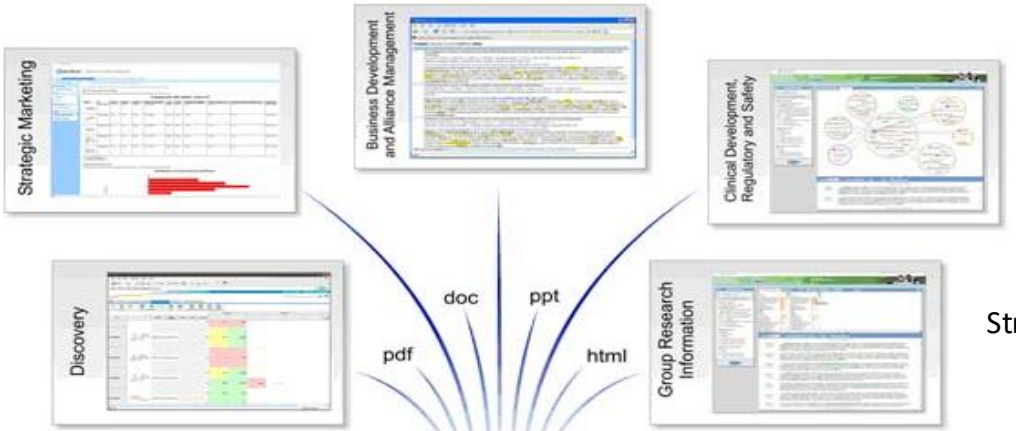


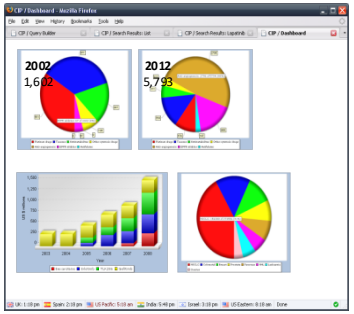
Trivialized view of a sample semantic solution architecture



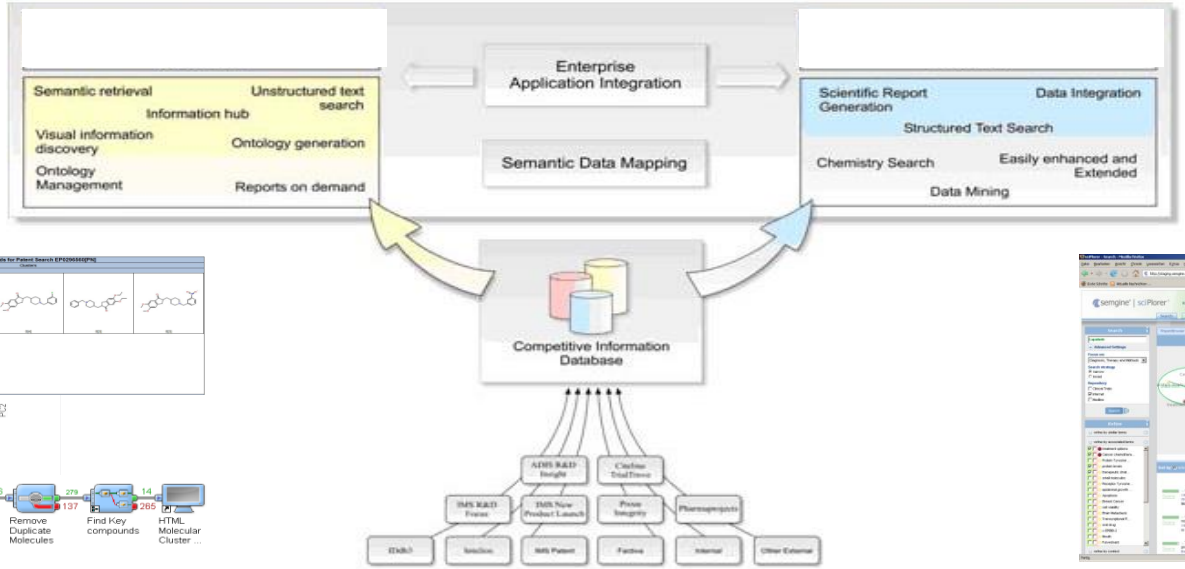


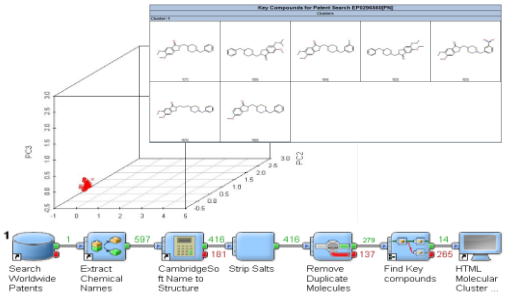
Detailed Product Compound Information



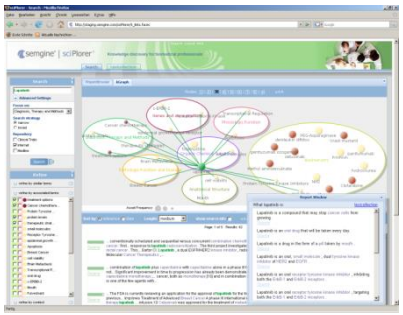


Strategic Information Dashboard





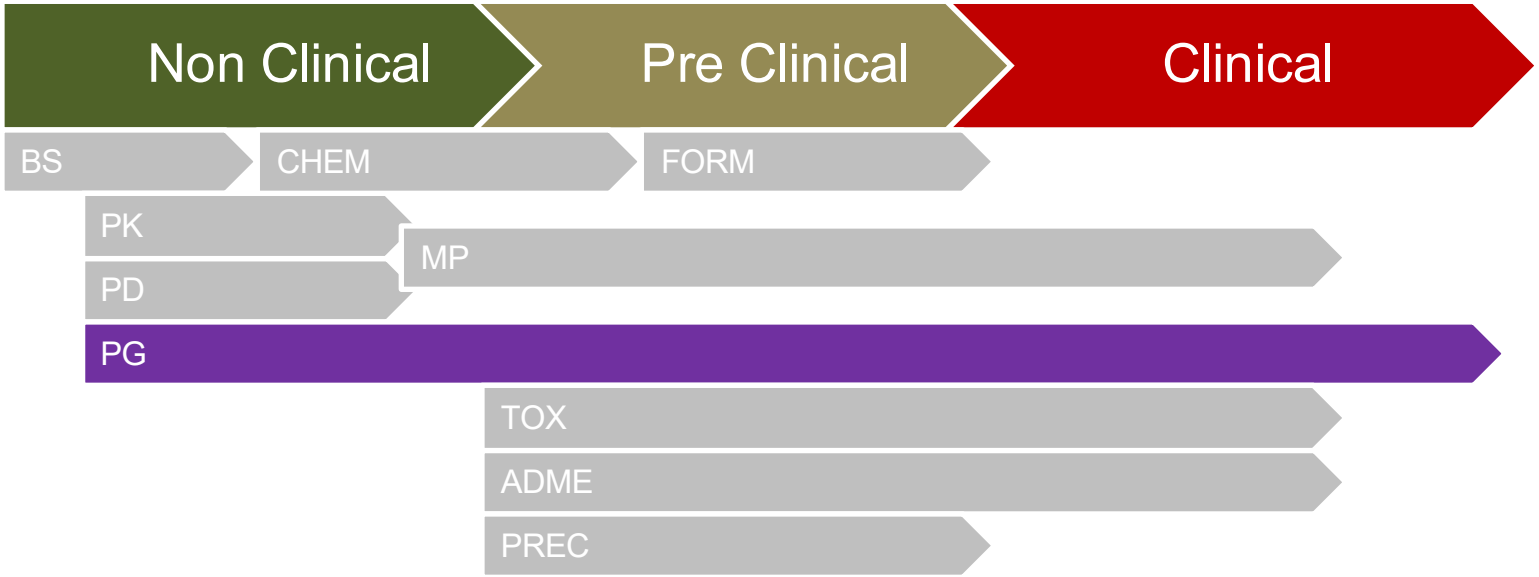
Chemistry, Stability, and Kinetics



Explore Semantically

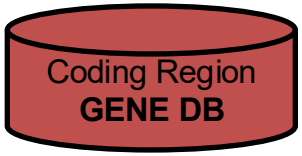
Key Data Sets in Pharmacogenomics

Pharmacogenomics has the longest need, staying power, and value of any set of research-based data



Pharmacogenomics

- Gene Defects or Mutations
- Multiple Allelic Variants
- Environmental Factors
- DNA Structure, Protein Structure
- Homology
- RNA and Ribosomal Targets
- Functional Genomics (expression)
- Regulatory Networks
- Protein-Protein Interactions
- Gene Knockouts Effects
- Gene Up Regulation Effects



- Disease Genes Identification
- Gene-Disease Mechanism
- Gene Expression High Density Arraying
- Gene Products Activity
- Bioinformatics Equipment Availability
- Interactions
- Pathways

BS = Basic Sciences
CHEM = Biochemistry
FORM = Formulation and Stability
PK/PD = Pharmaco Kinetics/Dynamics
MP = Molecular Profiling, and Biomarkers
PG = Pharmacogenomics
TOX = Toxicology
ADME = Absorption, Distribution, Metabolism, Excretion
PREC = Animal Sciences